

Ordinary Differential Eqn

$$y = f(x)$$

↓

$y' \rightarrow$ ordinary derivate.

$$y = f(x, y)$$

↓ y'

Partial differentiation.

→ Order :- highest ^{degree} power of diff. eqn.

$$\frac{d^3 y}{dx^3} + 3 \frac{dy}{dx} + xy = \left(\frac{d^2 y}{dx^2} \right)^2$$

Degree :- highest power of highest degree.

* Solⁿ of O.D.Eqn.

- i) Variable separable
- ii) Homogeneous Diff Eqn
- iii) Linear Diff Eqn.
- iv) Exact Diff Eqn.

§ Solve :-

$$(x+1) \frac{dy}{dx} = x(y^2+1).$$

$$\int \frac{dy}{y^2+1} = \int \frac{x}{x+1} dx$$

$$\Rightarrow \tan^{-1} y = \int \left(\frac{x+1}{x+1} - \frac{1}{x+1} \right) dx$$

$$\Rightarrow \tan^{-1} y = x - \log(x+1) + C$$

Q. Solve:-

$$(xy^2 + x) dx + (yx^2 + y) dy = 0$$

$$x(y^2 + 1) dx + y(x^2 + 1) dy = 0$$

$$x(y^2 + 1) dx = -y(x^2 + 1) dy$$

$$\frac{x}{x^2 + 1} dx = \frac{-y}{y^2 + 1} dy$$

$$\begin{aligned} x^2 + 1 &= t &= - \int \frac{y}{y^2 + 1} dy \\ 2x &= \frac{dt}{dx} \end{aligned}$$

$$x dx = \frac{dt}{2}$$

$$\begin{aligned} y^2 + 1 &= t \\ \Rightarrow \end{aligned}$$

$$\frac{1}{2} \int \frac{dt}{t}$$

$$\frac{1}{2} \log t = \frac{1}{2} \log t + C$$

$$\boxed{\frac{1}{2} \log(x^2 + 1) = \frac{1}{2} \log(y^2 + 1) + \frac{1}{2} \log C}$$

$$\Rightarrow \frac{(x^2 + 1)}{(y^2 + 1)} = C$$

Q. Solve:-

$$3e^x \tan y \, dx + (1 - e^x) \sec^2 y \, dy = 0.$$

$$\left| \begin{array}{l} e^x = t \\ dx = dt \end{array} \right.$$

$$\Rightarrow 3e^x \tan y \, dx = (e^x - 1) \sec^2 y \, dy.$$

$$\Rightarrow \frac{3e^x \, dx}{e^x - 1} = \int \frac{\sec^2 y \, dy}{\tan y}$$

$$\Rightarrow 3 \int \frac{dt}{t}$$

$$\left| \begin{array}{l} \tan y = t \\ dx \sec^2 y = dt \end{array} \right.$$

$$\Rightarrow 3 \log(e^x - 1) + \log c = \log(\tan y).$$

$$\Rightarrow \frac{(e^x - 1)^3}{\tan y} = c$$

Homogenous diff eqn

$$\frac{dy}{dx} = \phi(x, y)$$

$$\frac{dy}{dx} = f(x, y)$$

Put $x = vy$

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\frac{dy}{dx} = v + y \frac{dv}{dx}$$

Ex: Solve:-

$$x^2 dy + y(x+y) dx = 0$$

$$\frac{dy}{dx} = \frac{-y(x+y)}{x^2}$$

$y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} = \frac{-(vx)(x+vx)}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = -v - v^2$$

$$\Rightarrow 2v + v^2 = -x \frac{dv}{dx}$$

$$\Rightarrow \frac{dx}{x} = - \frac{dv}{2v + v^2}$$

$$\Rightarrow \int \frac{dx}{x} = - \int \frac{dv}{2v + v^2} = - \int \frac{dv}{v(v+2)}$$

$$\Rightarrow \frac{1}{2} \left(\frac{1}{v} - \frac{1}{v+2} \right) dv = - \frac{dx}{x}$$

$$\Rightarrow \log v - \log(v+2) = -2 \log x + \log c$$

$$\Rightarrow \log v - \log(v+2) + \log x^2 = \log c$$

$$\Rightarrow \log \left(\frac{v \cdot x^2}{v+2} \right) = \log c$$

$$\Rightarrow \frac{v \cdot x^2}{v+2} = c$$

$$\Rightarrow \frac{y \cdot x^2}{x} = c \Rightarrow \frac{y \cdot x^3}{y+2x} = c$$

$$\Rightarrow \frac{xy}{y+2x} = c \Rightarrow \frac{x^2 y}{y+2x} = c$$

Q. Solve:-

$$(x^3 - 3xy^2)dx + (y^3 - 3x^2y)dy = 0.$$

$$\frac{dy}{dx} = \frac{3x^2y - y^3}{x^3 - 3xy^2}$$

$$y = vx$$

$$v + x \frac{dv}{dx} = \frac{3x^2 \cdot vx - v^3 \cdot x^3}{x^3 - 3x \cdot v^2 x^2}$$

$$v + x \frac{dv}{dx} = \frac{3v - v^3}{1 - 3v^2}$$

$$x \frac{dv}{dx} = \frac{3v - v^3 - v + 3v^2}{1 - 3v^2}$$

$$x \frac{dv}{dx} = \frac{2v - v^3 + 3v^2}{1 - 3v^2}$$

$$x \frac{dv}{dx} = \frac{v^3 - 3v^2 - 2v}{(3v^2 - 1)}$$

$$\Rightarrow \frac{3v^2 - 1}{v^3 - 3v^2 - 2v} dv = \frac{dx}{x}$$

$$\Rightarrow \int \frac{3v^2 - 1}{v^3 - 3v^2 - 2v} dv = \int \frac{dx}{x}$$

$$3v^2 - 6v - 2 = \frac{d}{dv}$$

8 Solve $(x^3 - 3xy^2) dx + (y^3 - 3x^2y) dy = 0.$

$$\frac{dy}{dx} = \frac{x^3 - 3xy^2}{3x^2y - y^3}$$

$$v + x \frac{dv}{dx} = \frac{3v^2 - 1}{v^3 - 3v}$$

$$x \frac{dv}{dx} = \frac{-v^4 + 6v^2 - 1}{v^3 - 3v}$$

$$\Rightarrow \int \frac{v^3 - 3v}{v^4 - 6v^2 + 1} = - \int \frac{x}{dx}$$

$$\Rightarrow \frac{1}{4} \log(v^4 - 6v^2 + 1) = \log x + \log c$$

$$\Rightarrow \frac{(v^4 - 6v^2 + 1)^{1/4}}{x} = \underline{\underline{C}}$$

$$\Rightarrow \frac{\left(\left(\frac{y}{x} \right)^4 - 6 \left(\frac{y}{x} \right)^2 + 1 \right)^{1/4}}{x} = \underline{\underline{C}}$$

* Linear diff. Eqn :-

$$\frac{dy}{dx} + Py = Q.$$

$$I.F = e^{\int P dx}$$

Soln

$$y \cdot (I.F) = \int Q (I.F) dx.$$

Solve

$$(x+1) \frac{dy}{dx} - y = e^x (x+1)^2$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{(x+1)} = e^x (x+1).$$

$$I.F = e^{\int \frac{1}{x+1} dx} = e^{\log(x+1)}$$

$$= \frac{1}{(x+1)}.$$

Soln

$$\frac{y}{(x+1)} = \int e^x dx$$

$$\frac{y}{x+1} = e^x + C.$$

Q Solve:-

$$\frac{dy}{dx} + 2y \tan x = \sin x$$

$$\begin{aligned} \text{I.f} = e^{\int 2 \tan x \, dx} &= e^{\log |\sec x|^2} \\ &= \sec^2 x. \end{aligned}$$

$$\begin{aligned} \therefore \text{Sol}^n: y \cdot \sec^2 x &= \int \sin x \cdot \sec^2 x \, dx \\ &= \int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} \, dx \end{aligned}$$

$$y \sec^2 x = \sec x + C.$$

Q Solve:-

$$x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1.$$

$$\frac{dy}{dx} + y \frac{(x \sin x + \cos x)}{x \cos x} = \frac{1}{x \cos x}.$$

$$\begin{aligned} \text{I.f} = e^{\int \frac{x \sin x + \cos x}{x \cos x} \, dx} &= e^{\int \left(\tan x + \frac{1}{x} \right) dx} \\ &= e^{\log \sec x + \log x} \\ &= e^{\log x \sec x} \\ &= x \sec x. \end{aligned}$$

Soln

$$\int \sec u = \int \frac{1}{\cos u} \cdot \sec u \, du$$

$$= \int \sec u \, du$$

$$\boxed{\int \sec u = \ln |\sec u + \tan u| + C}$$

Exact differential Equation

→ An eqn of the form $Mdx + Ndy = 0$ is called exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution is given by

$$\int M dx + \int N dy = C$$

treating y as const. terms not containing x

→ Solve :-

$$(e^y + 1) \cos x \, dx + e^y \sin x \, dy = 0$$

$$M = (e^y + 1) \cos x$$

$$N = e^y \sin x$$

$$\frac{\partial M}{\partial y} = e^y \cos x$$

$$\frac{\partial N}{\partial x} = e^y \cos x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

→ the eqn is exact

$\therefore$ Soln is given by

$$\int (e^y + 1) \cos x \, dx + \int 0 \, dy = C$$

$$\Rightarrow (e^y + 1) \sin x = C$$

8) Solve

$$(5x^4 + 3x^2y^2 - 2xy^3) dx + (2x^3y - 3x^2y^2 - 5y^4) dy = 0$$

$$\Rightarrow M = 5x^4 + 3x^2y^2 - 2xy^3$$

$$N = 2x^3y - 3x^2y^2 - 5y^4$$

$$\frac{\partial M}{\partial y} = 6x^2y - 6xy^2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial N}{\partial x} = 6x^2y - 6xy^2$$

$\therefore$ This eqⁿ is exact.

$\therefore$ solⁿ is given by :-

$$\int (5x^4 + 3x^2y^2 - 2xy^3) dx - \int 5y^4 dy = c$$

$$= \cancel{20x^5} + 6xy^2$$

$$= x^5 + xy^2 - x^2y^3 - y^5 = c$$



Q Soln:

$$\underbrace{(x^3 - 3xy^2)}_M dx + \underbrace{(y^3 - 3x^2y)}_N dy = 0$$

$$\therefore \frac{\partial M}{\partial y} = -6xy$$

✓ This eqn is exact.

$$\frac{\partial N}{\partial x} = -6xy$$

$\therefore \phi(x, y)$ is given by

$$\int (x^3 - 3xy^2) dx + \int y^3 dy = C$$

$$\Rightarrow \frac{x^4}{4} - \frac{3x^2}{2}y^2 + \frac{y^4}{4} = C$$

Equation Reducible to Exact :-

↳ Rule 1 :-

if $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$, then

$$I.F = e^{\int f(x) dx}$$

Q. Solve :-

$$(2x \log x - xy) dy + \frac{2y}{x} dx = 0$$

Soln

→ This is not exact.

$$\frac{\partial M}{\partial y} = 2$$

$$\frac{\partial N}{\partial x} = 2 \log x + 2 - y$$

$$\therefore \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$$

$$= \frac{1}{(2x \log x - xy)} (2 \log x (2 - 2 \log x - 1 + y))$$

$$= -\frac{1}{x}$$

$$\therefore f(x) = -\frac{1}{x} \int \frac{1}{x} dx$$

$$\therefore I \cdot f = e^{\int \frac{1}{x} dx} = \frac{1}{x} \rightarrow \text{multiply to eq 1}$$

$$\Rightarrow (2 \log x - y) dy + \frac{2y}{x} dx = 0$$

$$\frac{\partial M}{\partial y} = 2$$

$$\frac{\partial N}{\partial x} = 2$$

∴ This is exact

Soln

$$\int \frac{2y}{x} dx - \int y dy = C$$

$$\Rightarrow 2y \log x - \frac{y^2}{2} = C$$

→ Rule 2:-

$$\text{if } \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = f(y)$$

$$\text{then I.F.} = e^{\int f(y) dy}$$

Solve:-

$$(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0.$$

Soln

$$\frac{\partial M}{\partial y} = 4y^3 + 2$$

$$\frac{\partial N}{\partial x} = 3xy^2 - 8y^3$$

$$\therefore \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = \frac{1}{(y^4 + 2y)} (3xy^2 - 8y^3 - 4y^3 - 2)$$

$$= \frac{-3y^3 - 6}{(y^4 + 2y)}$$

$$= -\frac{3}{y} = f(y).$$



$$\begin{aligned} I \cdot f &= e^{\int \frac{-3}{y} dy} = e^{-3 \log y} \\ &= e^{\log y^{-3}} \\ &= y^{-3} = \frac{1}{y^3} \end{aligned}$$

$$\therefore \frac{1}{y^3} (y^4 + 2y) dx + \frac{1}{y^3} (xy^3 + 2y^4 - 4x) dy = 0,$$

which is exact

Soln is

$$\Rightarrow \int \left(y + \frac{2x}{y^2} \right) dx + \int 2y dy = C$$

$$\Rightarrow \frac{y^2}{2} - \frac{x^2}{y^2} + y^2 = C$$
$$\Rightarrow \frac{3y^2}{2} - \frac{x^2}{y^2} = C$$

$$\Rightarrow xy + \frac{2x}{y^2} + y^2 = C$$

* Rule 3 :-

if $Mdx + Ndy = 0$ is a homogenous eqⁿ of x & y
then I.f = $\frac{1}{Mx + Ny}$

Q Solve :-

$$\frac{dy}{dx} = \frac{x^3 + y^3}{xy^2}$$

$$\Rightarrow (x^3 + y^3) dx = xy^2 dy$$

$$(x^3 + y^3) dx - xy^2 dy = 0$$

$$\Rightarrow M = x^3 + y^3$$

$$N = -xy^2$$

$$\therefore I.o.f = \frac{1}{x^4}$$

$$\therefore \frac{1}{x^4} (x^3 + y^3) dx - \frac{xy^2}{x^4} dy = 0$$

$$\left(\frac{1}{x} + \frac{y^3}{x^4} \right) dx - \frac{y^2}{x^3} dy = 0$$

$$\therefore \underline{\underline{\text{Soln}}} \int \left(\frac{1}{x} + \frac{y^3}{x^4} \right) dx + \int 0 dy = C$$

$$\boxed{\log x - \frac{y^3}{3x^3} = C}$$

1) Rule 4: if ~~Mdy~~ + Mdu + Ndy = 0 is of the form

$$y f_1(xy) du + u f_2(xy) dy = 0$$

$$\text{then } I.f = \frac{1}{M u - N y}$$

* Solve:

$$(xy^2 + 2x^2y^3) du + (x^2y - x^3y^2) dy = 0$$

$$\Rightarrow y(xy^2 + 2x^2y^3) du + x(x^2y - x^3y^2) dy = 0$$

$$\therefore I.f = \frac{1}{\cancel{x^2y^2} + 2x^3y^3 - \cancel{x^2y^2} - x^3y^3}$$

$$\Rightarrow = \frac{1}{3x^3y^3}$$

$$\therefore \frac{1}{3x^3y^3} (xy^2 + 2x^2y^3) du + \int \frac{(x^2y - x^3y^2)}{3x^3y^3} dy = 0$$

$$\Rightarrow \left(\frac{1}{3x^2y} + \frac{2}{3x} \right) du + \left(\frac{1}{3xy^2} - \frac{1}{3y} \right) dy$$

$$\therefore \int \left(\frac{1}{3x^2y} + \frac{2}{3x} \right) du = -\frac{1}{3y} \left(\frac{1}{x} \right) + \frac{2}{3} \log x$$

$$-2+1 = (-1)$$

$$\int -\frac{1}{3y} dy = -\frac{1}{3} \log y.$$

$$\therefore -\frac{1}{3y} + \frac{2}{3} \log x - \frac{1}{3} \log y = C.$$

→ Rule 5

$$I.F = x^h y^k$$

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

Q. Solve

$$(y^3 - 2x^2y) dx + (2xy^2 - x^3) dy = 0.$$

Soln

$$I.F = x^h y^k$$

$$\therefore x^h y^k [(y^3 - 2x^2y) dx + (2xy^2 - x^3) dy] = 0$$

$$(x^h y^{k+3} - 2x^{h+2} y^{k+1}) dx + (2x^{h+1} y^{k+2} - x^{h+3} y^k) dy = 0.$$

$$\frac{\partial M}{\partial y} = (k+3) x^h y^{k+2} - 2(h+2) x^{h+2} y^k$$

$$\frac{\partial N}{\partial x} = 2(h+1) x^h y^{k+2} - (h+3) x^{h+2} y^k$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\begin{aligned} 2x+2 &= h+3 \\ 2x &= h+1 \end{aligned}$$

$$\therefore (x+3) = 2(h+1)$$

$$f(2x+1) = f(h+3)$$

$$x+3 = 2h+2$$

$$x = 2h-1$$

$$2(2h-1)+1 = h+3$$

$$4h-2+1 = h+3$$

$$3h = 2$$

$$h = \frac{2}{3}$$

$$h = \frac{2}{3}$$

$$\begin{aligned} x &= 1 \\ h &= 1 \end{aligned}$$

$$\therefore \oint_C \vec{a} \cdot d\vec{r} = 0$$

$$\int (xy^4 - 2x^3y^2) dx + (2x^2y^3 - x^4y) dy = 0$$

$$\therefore \int (xy^4 - 2x^3y^2) dx = C$$

$$\frac{x^2y^4}{2} - \frac{2x^4}{4} y^2 = C$$

$$\left[\frac{x^2y^4}{2} - \frac{x^4}{2} y^2 = C \right]$$

→ Solve

$$y(x^2y + e^x) dx - e^x dy = 0$$

Solⁿ

← This is not Exact.

$$\hookrightarrow \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$$

$$\hookrightarrow \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = f(y)$$

$$\therefore \frac{\partial M}{\partial y} = 2x^2y + e^x$$

$$\frac{\partial N}{\partial x} = -e^x$$

$$\therefore \frac{1}{x^2y^2 + ye^x} \left(-e^x - 2x^2y + e^x \right)$$

$$= \frac{-2e^x - 2x^2y}{x^2y^2 + ye^x}$$

$$= \frac{-2}{y} \left(\frac{e^x + x^2y}{x^2y + e^x} \right) = \frac{-2}{y}$$

$$I.f = e^{\int -\frac{2}{y} dy} = e^{-2 \log y} = \frac{1}{y^2}$$

$$\Rightarrow \frac{1}{y^2} (x^2 y + e^x) dx - \frac{e^x}{y^2} dy$$

$$\left(x^2 + \frac{e^x}{y} \right) dx - \frac{e^x}{y^2} dy$$

$$\therefore \underline{\underline{\text{Sol}^n}} \quad \int \left(x^2 + \frac{e^x}{y} \right) dx = C$$

$$\boxed{\frac{x^3}{3} + \frac{e^x}{y} = C}$$

$$\oint (xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$$

$$\therefore \underline{\underline{I.f}} = \frac{1}{Mx - Ny}$$

$$= \frac{1}{x^2 y^2 \sin xy + xy \cos xy - \left(x^2 y^2 \sin x^2 - xy \cos x^2 \right)}$$

$$= \frac{1}{2xy \cos xy}$$

$$\therefore \frac{\oint (xy \sin xy + \cos xy) dx + x (xy \sin xy - \cos xy) dy}{2xy \cos xy}$$

$$= \left(\frac{y \sin xy}{2 \cos xy} + \frac{y}{2xy} \right) dx + \left(\frac{x}{2} \frac{\sin xy}{\cos xy} - \frac{x}{2xy} \right) dy = 0$$

$$\therefore \int \frac{y}{2} \frac{\sin xy}{\cos xy} dx + \frac{1}{2} \log x = \frac{1}{2} \log y$$

$$= \frac{y \log \sec xy}{2} + \frac{1}{2} \log x = \frac{1}{2} \log y$$

$$\Rightarrow \log(\sec xy)^{1/2}$$

$$\Rightarrow \frac{x \sec xy}{y} = C$$

$$\Rightarrow y \log y \, du + (n - \log y) \, dy = 0$$

$$m = y \log y$$

$$N = n - \log y$$

$$\frac{\partial m}{\partial y} = \log y + 1$$

$$\frac{\partial N}{\partial n} = 1$$

$$\therefore \frac{1}{N} \left(\frac{\partial m}{\partial y} - \frac{\partial N}{\partial n} \right) = f(n)$$

$$\left(\frac{1}{n - \log y} (\log y + 1 - 1) \right) = \frac{\log y}{n - \log y}$$

$$\frac{1}{m} \left(\frac{\partial N}{\partial n} - \frac{\partial m}{\partial y} \right) = f(y)$$

$$\left(\frac{1}{y \log y} (1 - \log y - 1) \right) = f(y)$$

$$= -\frac{1}{y}$$

$$\therefore -\log y \, du - \left(\frac{1}{y} - \frac{\log y}{y} \right) dy = 0$$

$$\therefore -\int \log y \, dy + \int \frac{\log y}{y} \, dy.$$

$$\begin{aligned} \log y &= t \\ \frac{1}{y} &= \frac{dt}{dy} \end{aligned}$$

$$-x \log y + \frac{\log^2 y}{2} = C$$

$$\begin{aligned} \int t \, dt \\ \frac{t^2}{2} \end{aligned}$$

* Linear differential Equation with constant coefficient :-

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = X.$$

where $a_1, a_2, \dots, a_n$ are constant & x is a f^n of x .

$$\frac{d}{dx} = D \quad ; \quad \frac{d^2}{dx^2} = D^2, \quad \frac{d^3}{dx^3} = D^3$$

$$\frac{d^n}{dx^n} = D^n$$

$$\Rightarrow D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \dots + a_{n-1} Dy + a_n y = X.$$

$$\Rightarrow (D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n) y = X$$

$$\Rightarrow f(D)y = X.$$

complete soln = complementary fn + Particular Integral.

if $x=0$, then $f(0)y = 0$

$$C \cdot s = C \cdot F$$

$$y = C \cdot F$$

complementary fn:-

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n) y = 0$$

Put $D = m$

$$m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0$$

↓
auxiliary Eqn

Case 1:- if roots are real & unequal.
 $m = m_1, m_2, m_3$

$$C \cdot F = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

Case 2:- if roots are real & equal.
let $m = m_1, m_1, m_1, m_2$

$$C \cdot F = (C_1 + C_2 x + C_3 x^2) e^{m_1 x} + C_4 e^{m_2 x}$$

Case 3:- if roots are complex & unequal.
let $m = \alpha_1 \pm i\beta_1, \alpha_2 \pm i\beta_2$

$$C \cdot F = e^{\alpha_1 x} (C_1 \cos \beta_1 x + C_2 \sin \beta_1 x) + e^{\alpha_2 x} (C_3 \cos \beta_2 x + C_4 \sin \beta_2 x)$$

Case IV if roots are complex & equal.

$$\text{Let } m = (\alpha \pm i\beta)^2$$

$$C.F. :- e^{\alpha x} [(C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x]$$

→ Q1 solve

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$$

$$D^2 y - 2Dy + y = 0$$

$$(D^2 - 2D + 1)y = 0$$

Put D=m

$$(m^2 - 2m + 1) = 0$$

$$m^2 - m - m + 1 = 0$$

$$(m-1)^2 = 0$$

$$m = 1, m = 1$$

$$\therefore C.F. = (C_1 + C_2 x) e^x$$

$$C.S. = (C_1 + C_2 e^x) e^x$$

Q2 Solve

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

Soln

$$D^2 y + 5Dy + 6y = 0$$

$$D = m$$

$$D^2 + 5D + 6 = 0$$

$$m^2 + 5m + 6 = 0$$

$$m^2 + 3m + 2m + 6 = 0$$

$$m(m+3) + 2(m+3) = 0$$

$$(m+3)(m+2) = 0$$

$$m = -2, -3$$

$$C.F. = c_1 e^{-2x} + c_2 e^{-3x}$$

Q3 $(D-2)^2 (D^2 + D + 1) y = 0$

$$(D^2 + 4D - 2D)(D^2 + D + 1) y = 0$$

$$(D^4 + D^3 + D^2 + 4D^2 + 4D + 4 - 4D^3 - 4D^2 - 4D) y = 0$$

$$(D^4 - 3D^3 + D^2 + 4) y = 0$$

$$(D^4 - 3D^3 + D^2 + 4) = 0$$

$$(D-2)^2 (D^2 + D + 1) y = 0$$

$$(m-2)^2 (m^2 + m + 1) = 0$$

$$\therefore m = 2, 2, \frac{-1 \pm \sqrt{3}i}{2}$$

$$C \cdot F = (C_1 + C_2 x) e^{2x} + e^{-\frac{x}{2}} \left(C_3 \cos \frac{\sqrt{3}x}{2} + C_4 \sin \frac{\sqrt{3}x}{2} \right)$$

Particular integral :-

$$f(D)y = x$$

$$p \cdot I = \frac{1}{f(D)} x$$

case 1 if $x = e^{ax}$

$$\text{then } p \cdot I = \frac{1}{f(D)} x$$

$$= \frac{1}{f(a)} x$$

[$\overset{\text{put}}{D=a}$] & $f(a) \neq 0$

$$= \frac{1}{f(a)} e^{ax}$$

if $f(a) = 0$

$$\text{then } p \cdot I = \frac{1}{f(D)} e^{ax}$$

$$= \frac{x}{f'(D)} e^{ax}$$

$$= \frac{x}{f'(a)} e^{ax} ; [D=a] ; f(a) \neq 0$$